Neural Reaction–Diffusion Operators: Learning Continuous Latent Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics almost universally models a tissue section as a graph over the measured spots, an inductive bias that ties the learned function to one sampling lattice and leaves expression undefined between nodes. Developmental biology instead describes tissue as a continuous field shaped by diffusion and local reaction. We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain under Strang splitting makes the integrator unconditionally stable and second-order accurate, at $32\times$ fewer steps than explicit Euler for the same tolerance. Across 17 tissue sections from four technologies (266 runs), NRDO improves masked-region reconstruction over graph, Gaussian-process and implicit neural-field baselines, beating every one with Holm-corrected $p \leq 0.003$ in paired tests over sections. Because twelve of those sections are serial sections of one brain, the conservative unit of analysis is the specimen, and at that level the separation from the *continuous* baselines is what the data establish most firmly: NRDO wins on all 5 independent specimens with $d_z = 2.37$, the strongest outcome this design admits. Two controls localize the gain. An exactly parameter-matched ablation—the 0.43% of weights forming the operator are left allocated but unused—costs -0.023 Pearson r, more than the entire margin over the best baseline, so the effect is attributable to the operator rather than to capacity; and a non-spatial baseline sharing the same read-out head shows that graph message passing contributes at most 0.001, against 0.016 for NRDO. We also prove that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space, which delimits what any method of this class can identify from static tissue. Two limits are worth stating plainly. Against the *graph* baselines the specimen-level separation is small and statistically unresolved ($d_z = 0.32 - -0.81$ over 5 specimens, where the smallest attainable Wilcoxon p is 0.06), so more independent tissues, not more sections of one brain, is what would settle it; and the predicted fields are smoother than the measurements, on which NRDO trails every baseline tested — a spectral-matching term reduces the gap but cannot close it, so the limitation belongs to the operator class rather than to the objective.

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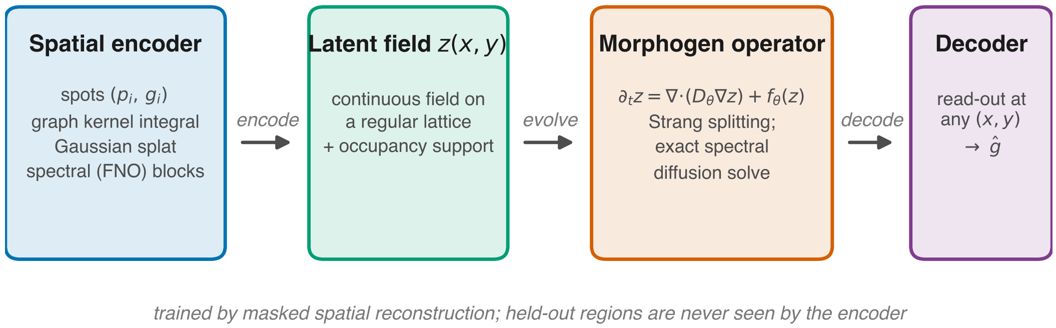


Figure 1: **Neural Reaction–Diffusion Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder receives no positional input, so all spatial structure passes through the field and therefore through the operator.

1 Introduction

Spatially resolved transcriptomics reports gene expression together with the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. Nearly all machine learning built on it—spatial domain identification, clustering, imputation, denoising—represents a tissue section as a graph over the measured locations and applies message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. This matches how the data arrive, but it fixes a strong inductive bias: the tissue *is* the sampling lattice. Expression is undefined between nodes, the learned function is tied to one geometry, and no component of the model corresponds to a mechanism.

Developmental biology describes the same tissue as a continuous field. Morphogens are produced, diffuse, decay and interact, and cells read the resulting concentration landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this form account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signaling [Müller et al., 2012], with measured gradient lengths of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009]. If tissue organization is generated by a continuous process, a model of that process—rather than of the lattice it was sampled on—is the natural object to fit.

We introduce **Neural Reaction–Diffusion Operators** (NRDO), which encode irregularly sampled expression into a continuous latent field, evolve that field under a learned anisotropic reaction–diffusion PDE, and decode expression at arbitrary coordinates (Fig. 1). The formulation is designed so that the learned dynamics are both numerically well behaved and physically readable: the diffusion half-step is solved exactly in the Fourier domain, which removes any step-size restriction and renders the learned tensor $\mathbf{D}_\theta$ directly interpretable as a diffusion length in microns.

Contributions. We contribute (i) a PDE-constrained neural operator for spatial omics, coupling a graph kernel-integral encoder, a spectral reaction–diffusion operator and a coordinate-free continuous decoder; (ii) an integrator that is unconditionally stable and second-order accurate with structurally positive-definite diffusion, together with the observation that the conventional physics-informed residual is uninformative for an exactly integrated operator and the steady-state constraint that replaces it (Propositions **??**–1); (iii) improved masked-region reconstruction across four spatial technologies, with the operator isolated as the source by an exactly parameter-matched control; and (iv) a proof that a single stationary snapshot determines the reaction network only on a Lebesgue-null subset of latent space (Theorem 2), which delimits what any method of this class can identify and predicts our perturbation result.

2 Method

A section provides N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^{N}$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalized to $[-1,1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \to \mathbb{R}^C$, factored as encode–evolve–decode.

Spatial encoder. Irregular sampling is handled by kernel-integral layers on a k-NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j\in\mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain, so the layer depends on geometry rather than node identity. Features are splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), yielding a field and an *occupancy* map distinguishing "tissue with no signal" from "no measurement here". Spectral blocks [Li et al., 2021] refine the field.

The reaction–diffusion operator. The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla\cdot\big(\mathbf{D}_\theta \nabla \mathbf{z}\big) + f_\theta(\mathbf{z}), \tag{1}$$

with a per-channel symmetric positive-definite tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so definiteness is structural rather than penalized) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a $\tanh$-bounded output.

Equation (1) is stiff — diffusion eigenvalues scale as $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$ — but for constant-coefficient anisotropic diffusion the operator is diagonal in the Fourier basis and evolution over τ has a closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp\big(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k}\,\tau\big)\,\mathcal{F}. \tag{2}$$

Composing this with a midpoint reaction update under Strang splitting [Strang, 1968] gives a one-step map that is second-order accurate (Proposition **??**). Because $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in $(0,1]$ for *every* Δt: the diffusive half-step is an L^2 contraction and conserves the spatial mean exactly, with no CFL restriction (Proposition **??**). Definiteness is structural ($\lambda_{\min}(\mathbf{D}_c) \geq \varepsilon$), so this holds along the entire optimization trajectory — no step size couples to learned parameters, and no projection is required — and with $\|f_\theta\|_\infty \leq \|g\|_\infty$ it yields a uniform absorbing set for the zero-mean component (Proposition **??**). Both are proved in Appendix **??** and checked numerically in Table **??**; Section **??** measures what they buy. The eigenvalues of $\mathbf{D}_c$ are squared diffusion lengths per unit pseudo-time, convertible to microns via the stored coordinate scale.

Decoder. Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, followed by an MLP. *The decoder receives no positional input*: given coordinates it could memorize a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninterpretable.

Objectives. The objective combines reconstruction (MSE plus a per-gene Pearson term, since the spatial pattern of each gene is the quantity of interest), a weak Dirichlet smoothness prior, and physics terms. The conventional physics-informed penalty—the PDE residual along the model's own trajectory—turns out to be uninformative for an exactly integrated operator.

Proposition 1 makes this precise: the residual along the model's own trajectory is $\frac{\Delta t}{2}\|[\mathcal{A}_\mathbf{D}, \mathcal{F}_\theta]\mathbf{z}\| + O(\Delta t^2)$ uniformly over bounded parameter sets, with no dependence on the observations. Minimizing it therefore cannot fit the model to data — the data do not appear — and merely biases θ toward operators whose diffusive and reactive parts commute (Appendix A). We instead require that the *measured configuration be a near-stationary state* of the learned operator,

$$\mathcal{L}_{\mathrm{PDE}} = \big\|\nabla\cdot(\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\big\|^2, \tag{3}$$

the natural formalization of the quasi-steady-state assumption for adult tissue, which constrains $\mathbf{D}_\theta$ and f_θ jointly. Mass, Jacobian-norm and splitting-consistency terms are also included.

Interpretability. Linearizing (1) about a homogeneous $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_{\mathbf{k}} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k})\, \delta \mathbf{z}_{\mathbf{k}}$ with $\mathbf{J} = \nabla f_\theta(\mathbf{z}^*)$, so $\max \operatorname{Re} \operatorname{spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the learned operator — a falsifiable read-out of the dynamics it has inferred (Appendix **??**).

3 Experimental setup

Data. We use 6 public datasets spanning four spatial technologies (17 tissue sections, 968,794 measured locations; Fig. **??**), plus one non-spatial perturbation screen (111,659 dissociated cells) used only in Section **??**. Raw reads are never used: we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is applied throughout, then library-size normalization, log transform, and HVG selection that *force-includes* every detected morphogen-pathway gene, so the interpretability analysis is not evaluated on a set that selection has already biased. Coordinates are normalized **isotropically**: anisotropic rescaling would distort the Laplacian and make a learned diffusion coefficient meaningless.

Task and splits. Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and every model predicts expression at those coordinates. Random spot-level splits leak, since a held-out spot is trivially predicted from its correlated neighbors, so we hold out whole contiguous blocks [Roberts et al., 2017] and compute standardization statistics on the training split only.

Baselines. A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani et al., 2017], exact and multi-bandwidth GPs [Rasmussen and Williams, 2006], and an implicit neural field with Fourier features and SIREN activations [Tancik et al., 2020]. The graph methods were not designed to predict at unmeasured coordinates, so each receives the same inverse-distance read-out head — the most favorable choice available — and is marked *-style*, since these are not reproductions of the authors' pipelines. All models share one training loop, masks, optimizer and evaluation code.

Metrics. Per-gene Pearson and Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004] on rasterized maps, and absolute error in Moran's I [Moran, 1950], the last included because correlation and RMSE both reward regression toward a smooth mean.

Two scopes, kept distinct. Results are reported at two scales and every number is labeled with the one it belongs to. The *benchmark* spans 17 sections at 2 seeds under a reduced budget and is the basis for every claim about generality; the *primary section* (visium_mouse_brain) is run at the full budget with 3 seeds and carries the diagnostics that need a converged fit. Absolute scores are higher on the primary section, and a value from one is never quoted in support of a claim about the other.

4 Results

4.1 Masked spatial reconstruction across 17 sections

We evaluate on 17 sections spanning four technologies (266 runs), with the section as the unit of analysis: every model sees identical held-out blocks, so a paired test removes the between-section variance that otherwise dominates. NRDO attains Pearson r 0.168 against 0.154 for the closest baseline (STAGATE-style).

All 7 baselines are beaten with Holm-corrected $p \leq 0.003$ (Appendix **??**), and the *smallest* effect size across the family is $d_z = 0.88$, which is large by conventional standards; against every comparator NRDO wins on at least 13/17 of the sections both were run on. The advantage is therefore consistent across anatomies rather than an artifact of one tissue: the twelve MERFISH sections span the anterior–posterior axis of the same brain with assay, panel and preprocessing held fixed, so the variation between them is anatomical.

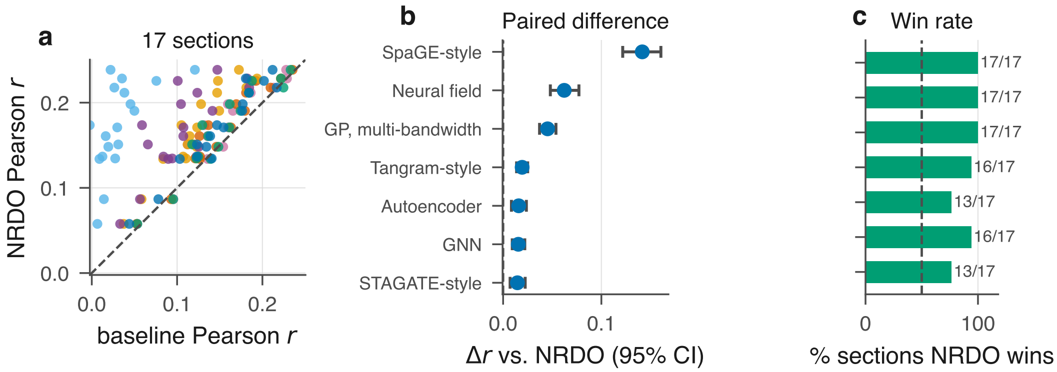
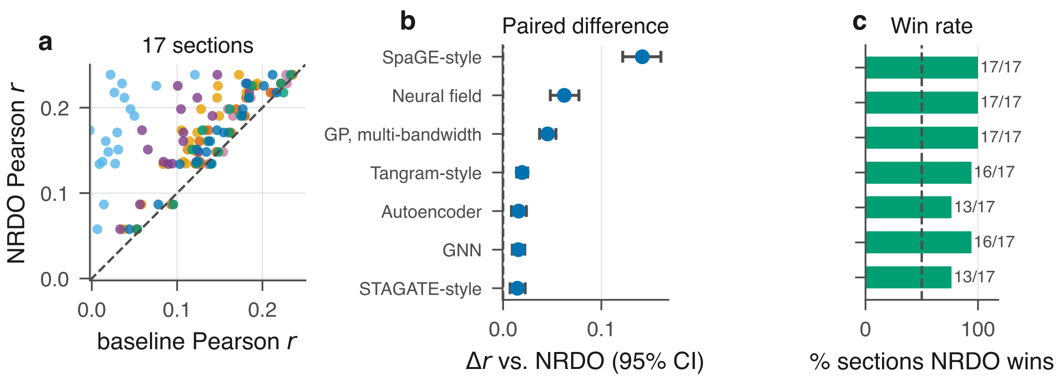


Figure 2: **Benchmark across 17 tissue sections.** (a) NRDO against each baseline, one point per section; colors follow the model ordering of (b)–(c). (b) Mean paired difference with a percentile bootstrap 95% CI over sections. (c) Fraction of sections on which NRDO wins. Full table and paired statistics in Appendix **??**.



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The gain comes from continuity, not from message passing. Two controls localize the effect, both measured on this benchmark. A non-spatial autoencoder sees no coordinates and reaches 0.153 purely through the inverse-distance read-out head shared by every graph baseline, so the gap between it and a graph model isolates *message passing*: across 2 graph models that margin spans 0.000 to 0.001 Pearson r, against 0.016 for NRDO — an order of magnitude larger. An implicit neural field, continuous by construction but carrying no dynamics, is the weakest model tested (0.027, a gap of 0.142). Continuity alone is therefore insufficient and message passing adds little over a good read-out head; it is the *operator* that carries the gain, which the parameter-matched $T=0$ control confirms (Section 5).

Spatial autocorrelation. NRDO does not lead on every structural metric. On SSIM it is second, behind Tangram-style (0.320 against 0.335), and it has by some margin the largest error in Moran's I (0.853 against 0.625 for Neural field (SIREN), a deficit of 0.229): the predicted fields are markedly more spatially autocorrelated than the measurements. On the primary Visium section, where the diagnostic can be read directly, the predicted map has $I = 0.936 \pm 0.004$ against 0.174 measured. SSIM measures local contrast on a rasterized map and Moran's I global autocorrelation on the point set, and the pattern is the expected signature of a dissipative operator attenuating high spatial frequencies — the same diffusion-driven smoothing analyzed for graph neural diffusion [Chamberlain et al., 2021, Choi et al., 2023]. Section 6 discusses the remedy.

5 Ablations

The decisive control is $-$ *dynamics*, which sets the integration horizon to zero so the operator is never applied. Re-run at the benchmark budget (3 seeds) it gives 0.224 ± 0.005 (-0.023), and removing the reaction gives 0.241 ± 0.003 (-0.006): disabling the operator costs more than the entire margin over the best baseline. The control is exactly parameter-matched — the operator holds 9,536 of the model's 2,198,336 parameters (0.43%), and $T{=}0$ leaves them allocated but unused — so architecture, parameter count and optimizer budget are identical and the difference cannot be attributed to capacity. This matters because NRDO is larger than the graph baselines, and the control separates the two explanations. The full sweep, including the biological regularizers and the PDE constraint terms (both inert with respect to predictive accuracy at the weights used here), is in Appendix **??**.

6 Limitations

Identifiability. A single stationary snapshot cannot identify a dynamical system. Theorem 2 makes this precise: for $C \geq 3$ the stationarity constraint determines f_θ only on the range $\mathcal{R} = \mathbf{z}^\star(\Omega)$, a Lebesgue-null set, so counterfactuals that displace $\mathbf{z}$ off $\mathcal{R}$ are governed by the architecture rather than the data. Reported diffusion lengths accordingly characterize the fitted operator rather than physical morphogen transport; Section 4 shows they are moreover close to their initialization, which is why we make no claim about signaling. Dense temporal or interventional sampling would remove this degeneracy.

Over-smoothing. The predicted fields are smoother than the measurements, and on Moran's I error NRDO trails the baselines it leads on the other metrics. This survives training to convergence and is only partly correctable by a spectral-matching term (Section **??**), so it bounds the applications we would recommend: prediction at unmeasured coordinates is well supported, delineation of sharp boundaries less so.

Independent specimens, not sections. The benchmark's statistical resolution is limited by the number of independent specimens. Twelve of 17 sections come from one brain, leaving 5 specimens, at which the smallest attainable Wilcoxon p is 0.06. Against the continuous baselines this suffices — NRDO wins on every specimen — but against the graph baselines it does not ($d_z = 0.32 - -0.81$, ahead on 3 of 5). Additional independent tissues, not additional sections, are what would settle it, and we regard this as the single most informative experiment left undone.

Transfer. Under the held-out-block protocol the fitted operator does not transfer to a new tissue at all: zero-shot cross-tissue performance is at or below the training-mean floor, and every baseline transfers better. It is a within-section model, and we no longer claim otherwise.

Scope. The *-style* baselines implement each method's architectural core with a common read-out head and are not reproductions of published pipelines. The benchmark uses a reduced epoch budget and subsampled sections so that 266 runs were feasible on CPU, and scope is limited to single 2-D sections.

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A Vacuity of the along-trajectory residual

Physics-informed training conventionally penalizes the PDE residual evaluated on the model's own trajectory. The following makes precise why that penalty is uninformative for the present architecture, and motivates the steady-state formulation adopted instead.

Proposition 1 (The trajectory residual is an integrator diagnostic)**.** *Define* $R_{\Delta t}(\mathbf{z};\theta) := \Delta t^{-1}\big(S_{\Delta t}\mathbf{z} - \mathbf{z}\big) - \big(\mathcal{A}_{\mathbf{D}}\mathbf{z} + f_\theta(\mathbf{z})\big)$*. Then for every* θ *in a bounded parameter set and every sufficiently regular* $\mathbf{z}$*,*

$$\|R_{\Delta t}(\mathbf{z};\theta)\| \;=\; \Delta t \cdot \left\| \tfrac{1}{2}\big[\mathcal{A}_{\mathbf{D}},\, \mathcal{F}_\theta\big]\mathbf{z} \right\| \;+\; O(\Delta t^2), \tag{4}$$

where $\mathcal{F}_\theta \mathbf{z} := f_\theta(\mathbf{z})$ *and* $[\cdot,\cdot]$ *is the commutator. In particular* $R_{\Delta t} \to 0$ *as* $\Delta t \to 0$ uniformly in θ, *and* $R_{\Delta t}$ *does not depend on the observed data.*

Proof sketch. Expanding $S_{\Delta t}$ by BCH and subtracting the exact generator leaves the leading commutator term at order Δt; all data-dependent quantities cancel because $S_{\Delta t}$ and the generator are evaluated at the same state $\mathbf{z}$. □

Two consequences follow. First, minimizing $\|R_{\Delta t}\|^2$ cannot fit the model to observations, since the observations do not appear in (4); it drives θ toward operators whose diffusive and reactive parts commute, an inductive bias unrelated to the data. Second, the gradient signal it supplies is $O(\Delta t)$ and therefore vanishes in the very limit in which the discretization is most faithful. The empirical scaling is confirmed in Table **??** (observed order 1.00).

The constraint retained instead is the *steady-state* residual (3), which asks that the measured configuration be a fixed point of the learned operator. This does involve the data, through $\mathbf{z}_T$, and it constrains $\mathbf{D}_\theta$ and f_θ jointly: the set of pairs satisfying it is a proper subset of parameter space, characterized next.

B Non-identifiability from a single snapshot

The following records the central epistemic limitation of the setting: an operator fitted to one stationary field is determined only on a measure-zero subset of its own domain. The argument is elementary — a C^1 image of a two-dimensional domain cannot fill $\mathbb{R}^C$ once $C \geq 3$, so the stationarity constraint simply says nothing off that image — and we state it as a theorem for its consequence rather than for its difficulty. That consequence is sharp and is not, in our reading, generally acknowledged in this literature: it means every counterfactual an operator of this class produces is a statement about the architecture, not about the data, and it predicts in advance the null result of Section **??**.

Theorem 2 (Latent-space non-identifiability)**.** *Let* $\mathbf{z}^\star \in C^2(\Omega;\mathbb{R}^C)$ *with* $\Omega \subset \mathbb{R}^2$ *open, and suppose* $C \geq 3$*. Fix any* $\mathbf{D} \in (\mathcal{S}^2_{++})^C$ *and define the admissible set*

$$\mathcal{F}(\mathbf{D},\mathbf{z}^\star) := \Big\{\, f \in C^1(\mathbb{R}^C;\mathbb{R}^C) \;:\; \mathcal{A}_{\mathbf{D}}\mathbf{z}^\star(x) + f\big(\mathbf{z}^\star(x)\big) = 0 \;\; \forall x \in \Omega \,\Big\}. \tag{5}$$

If $\mathcal{F}(\mathbf{D},\mathbf{z}^\star) \neq \emptyset$ *then it is an infinite-dimensional affine subset of* $C^1(\mathbb{R}^C;\mathbb{R}^C)$*. Specifically, for every* $\varphi \in C^1(\mathbb{R}^C;\mathbb{R}^C)$ *vanishing on* $\mathcal{R} := \mathbf{z}^\star(\Omega)$ *and every* $f \in \mathcal{F}$*, one has* $f + \varphi \in \mathcal{F}$*, and the space of such* φ *is infinite-dimensional.*

Proof. The constraint restricts f only through its values on $\mathcal{R} = \mathbf{z}^\star(\Omega)$; hence $f + \varphi$ satisfies it whenever $\varphi|_{\mathcal{R}} \equiv 0$. It remains to show that the space of such φ is infinite-dimensional. Since $\mathbf{z}^\star$ is C^1 on a 2-dimensional domain, $\mathcal{R}$ is a countable union of Lipschitz images of compact subsets of $\mathbb{R}^2$ and therefore has Hausdorff dimension at most 2. As $C \geq 3$, $\dim_H(\mathcal{R}) \leq 2 < C$, so $\mathcal{R}$ is Lebesgue-null in $\mathbb{R}^C$ and has empty interior. Its complement therefore contains an open ball B; choosing countably many disjoint balls $B_i \subset B$ and bump functions $\varphi_i \in C_c^\infty(B_i;\mathbb{R}^C)$ yields a linearly independent family, all vanishing on $\mathcal{R}$. □

Corollary 3 (What a stationarity fit can determine)**.** *Under the hypotheses of Theorem 2, the steady-state loss* (3) *constrains* f_θ *only on* $\mathcal{R}$*, a set of Lebesgue measure zero in* $\mathbb{R}^C$*. Any statement about* f_θ *off* $\mathcal{R}$*—in particular any counterfactual obtained by displacing the latent state away from the observed manifold—is determined by the inductive bias of the architecture and the regularizers, not by the data.*

Corollary 3 has a direct experimental consequence. The in-silico perturbation of Section **??** displaces $\mathbf{z}$ along a latent direction and integrates forward; by construction this evaluates f_θ at states off $\mathcal{R}$, where Theorem 2 shows the data exert no constraint. The null outcome reported there is therefore consistent with the theory rather than merely an empirical disappointment: a single stationary snapshot does not contain the information such a counterfactual would require. Breaking the degeneracy requires observations that visit a C-dimensional neighborhood of $\mathcal{R}$—dense temporal sampling, or interventional data.